Module 4

Planning and configuring directory synchronization

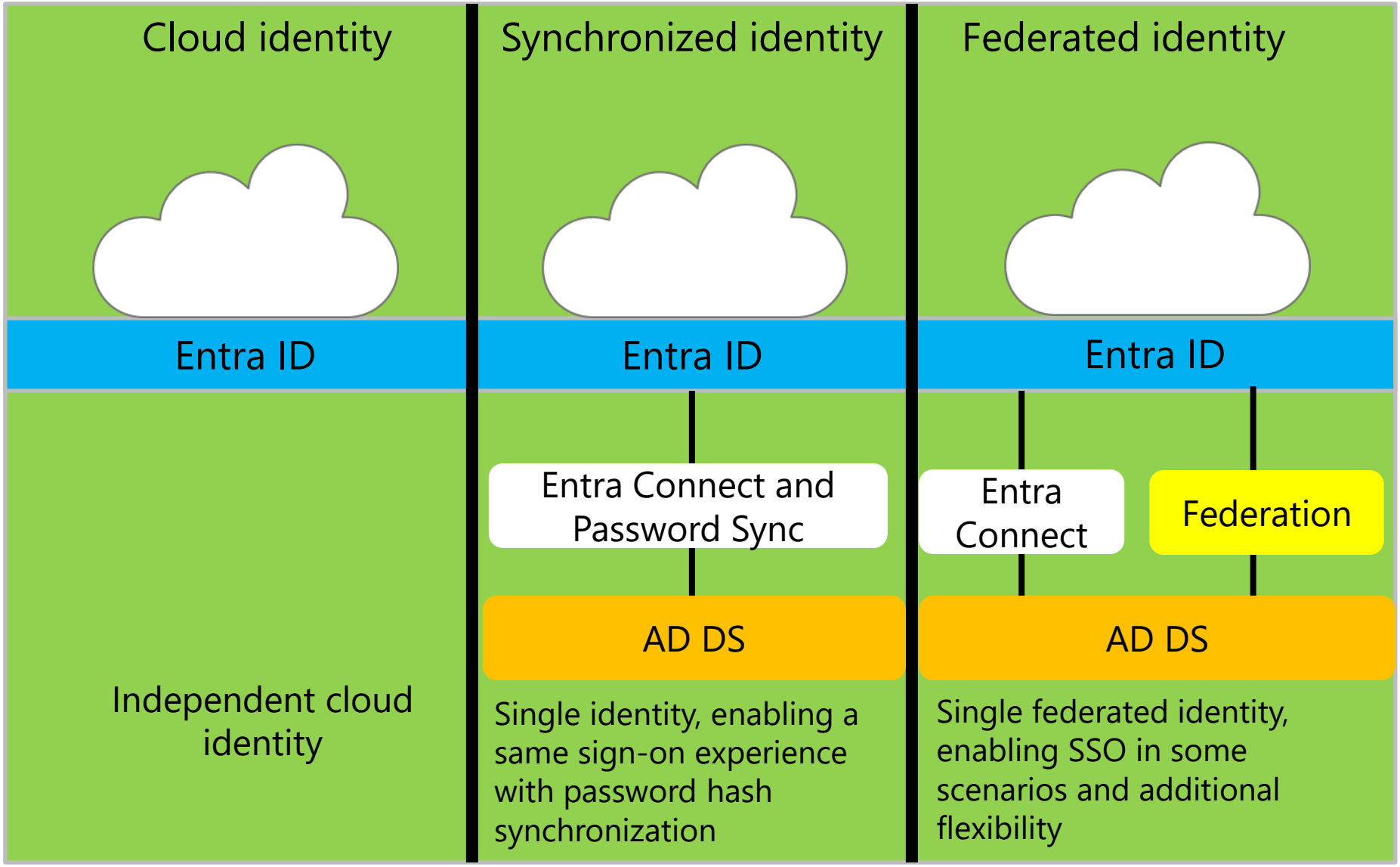
Module Overview

- Planning and preparing for directory synchronization
- Implementing directory synchronization by using Entra Connect
- Managing Microsoft 365 identities with directory synchronization

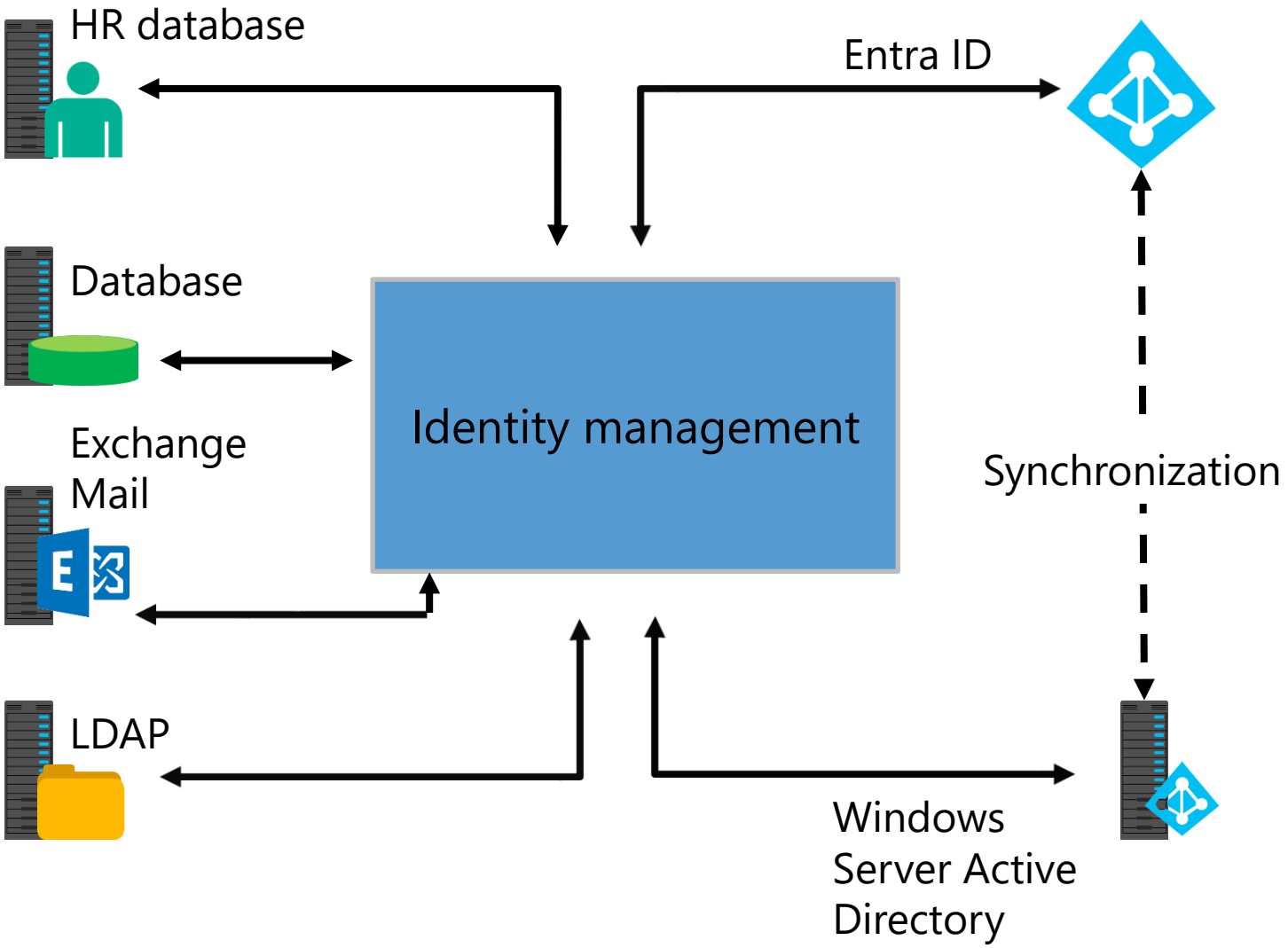
Lesson 1: Planning and preparing for directory synchronization

- Microsoft 365 authentication options
- Overview of directory synchronization
- Planning directory synchronization
- Prerequisites for directory synchronization
- Preparing for directory synchronization
- Configuring a tenant for directory synchronization

Microsoft 365 authentication options



Overview of directory synchronization



Planning directory synchronization

Best practices for deploying directory synchronization:

- Have a proper project plan
- If AD DS filtering is used, configure it before synchronizing objects to Microsoft 365
- Work with a cloud services partner
- Perform thorough capacity planning
- Remediate AD DS before deploying directory synchronization
- Add all SMTP domains as verified domains before synchronizing

Prerequisites for directory synchronization

When reviewing the prerequisites for directory synchronization, your tasks should include:

- Capacity planning for your directory synchronization database server
- Identifying the hardware requirements for your directory synchronization computer
- Identifying if your environment exceeds the Entra ID object quota
- Reviewing the network ports required by directory synchronization
- Determining if any schema extensions to AD DS are required

Preparing for directory synchronization

You will need to prepare your environment for directory synchronization

- If you will be using SSO in your environment, then you should deploy it before directory synchronization
- You will need to prepare your on-premises AD DS environment
- You will identify and configure the appropriate UPN suffixes in your on-premises AD DS environment
- You will use Microsoft 365 IdFix to resolve any issues identified by the Microsoft 365 readiness checks

Configuring a tenant for directory synchronization

- To enable Active Directory synchronization by using the Microsoft 365 admin center you should start Directory sync setup assistant
- You can also enable directory synchronization by using Windows PowerShell
- You can view status of the directory synchronization by using DirSync Status in the Microsoft 365 admin center

Lesson 2: Implementing directory synchronization by using Entra Connect

- Overview of Entra Connect
- Entra Connect requirements
- Entra Connect express synchronization settings
- Entra Connect customized synchronization
- Upgrading to Entra Connect
- Entra Connect Health
- Entra Connect in multi-forest scenarios
- Entra Connect pass-through authentication

Overview of Entra Connect

When using Entra Connect for directory synchronization:

- New user, group, and contact objects in on-premises AD DS are added to Microsoft 365
- Attributes of existing user, group, or contact objects that are modified in on-premises AD DS are modified in Microsoft 365
- Existing user, group, and contact objects that are deleted from on-premises AD DS are deleted from Microsoft 365
- Existing user objects that are disabled on-premises are disabled in Microsoft 365

Entra Connect requirements

When identifying the Entra Connect requirements, you should assess:

- Entra ID requirements
- Domain and forest requirements
- Operating system and supporting software requirements
- Permissions and accounts
- Database requirements

Entra Connect express synchronization settings

- Scenarios for using the Express Settings include:
  - If you have a single forest AD DS
  - Users sign in with the same password using password synchronization
- Installing Entra Connect with Express Settings:
  - Installs the synchronization engine
  - Configures Entra Connector
  - Configures the on-premises AD DS connector
  - Enables password synchronization
  - Configures synchronization services
  - Configures sync services for Exchange hybrid deployment (optional)
  - Enables automatic update for Entra Connect

Entra Connect customized synchronization

The scenarios for when to select customized settings include:

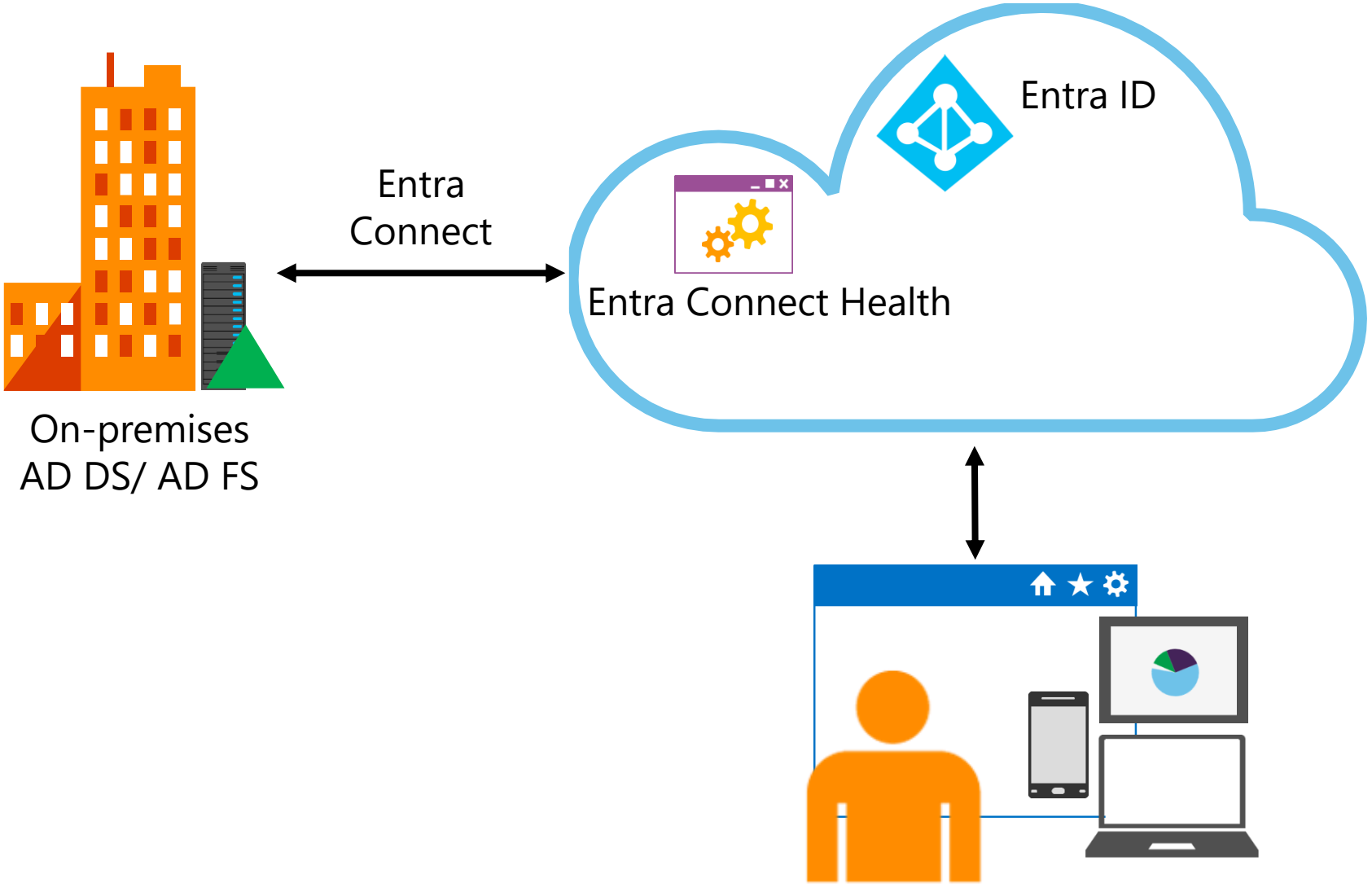
- When you have multiple forests
- When you customize your sign-in option, such as AD FS for federation or use a non-Microsoft identity provider
- When you customize synchronization features, such as filtering and writeback

Upgrading to Entra Connect

Different options for an Entra Connect upgrade:

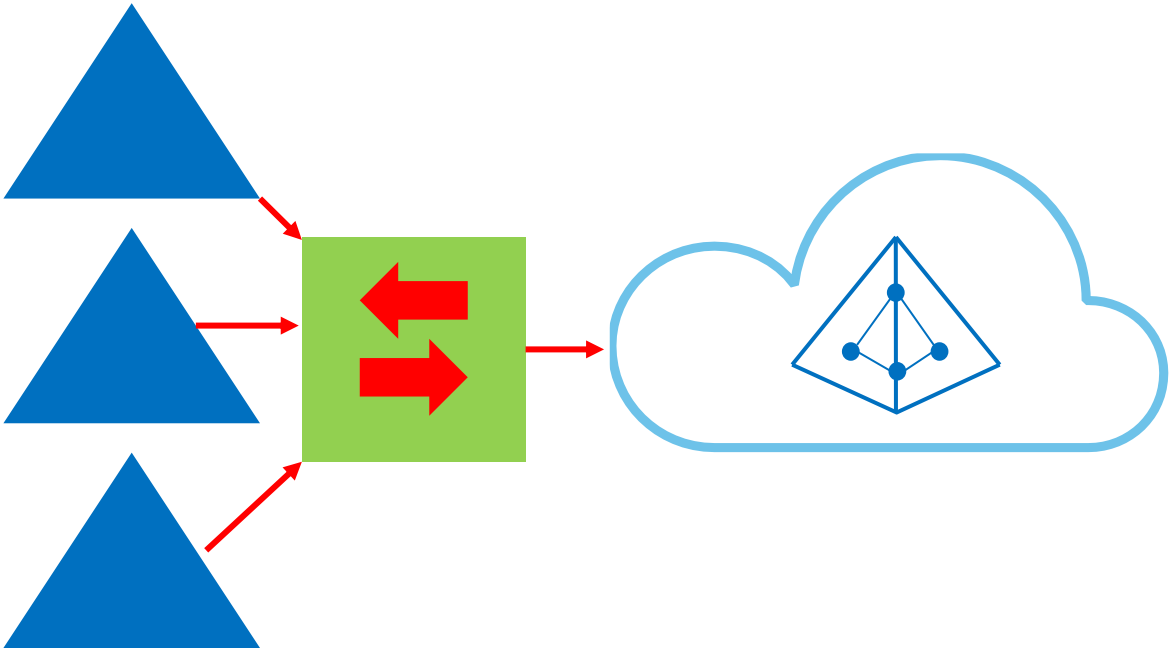
- In-place upgrade, if the expected upgrade time is less than three hours then the recommend option is to do an in-place upgrade
- Parallel deployment, if the expected upgrade time is more than three hours then the recommend option is to do a parallel deployment on another server. It is estimated that if you have more than 50,000 objects in AD DS, then it will take more than three hours to do the upgrade; the preferred upgrade option is a parallel deployment in this scenario

Entra Connect Health



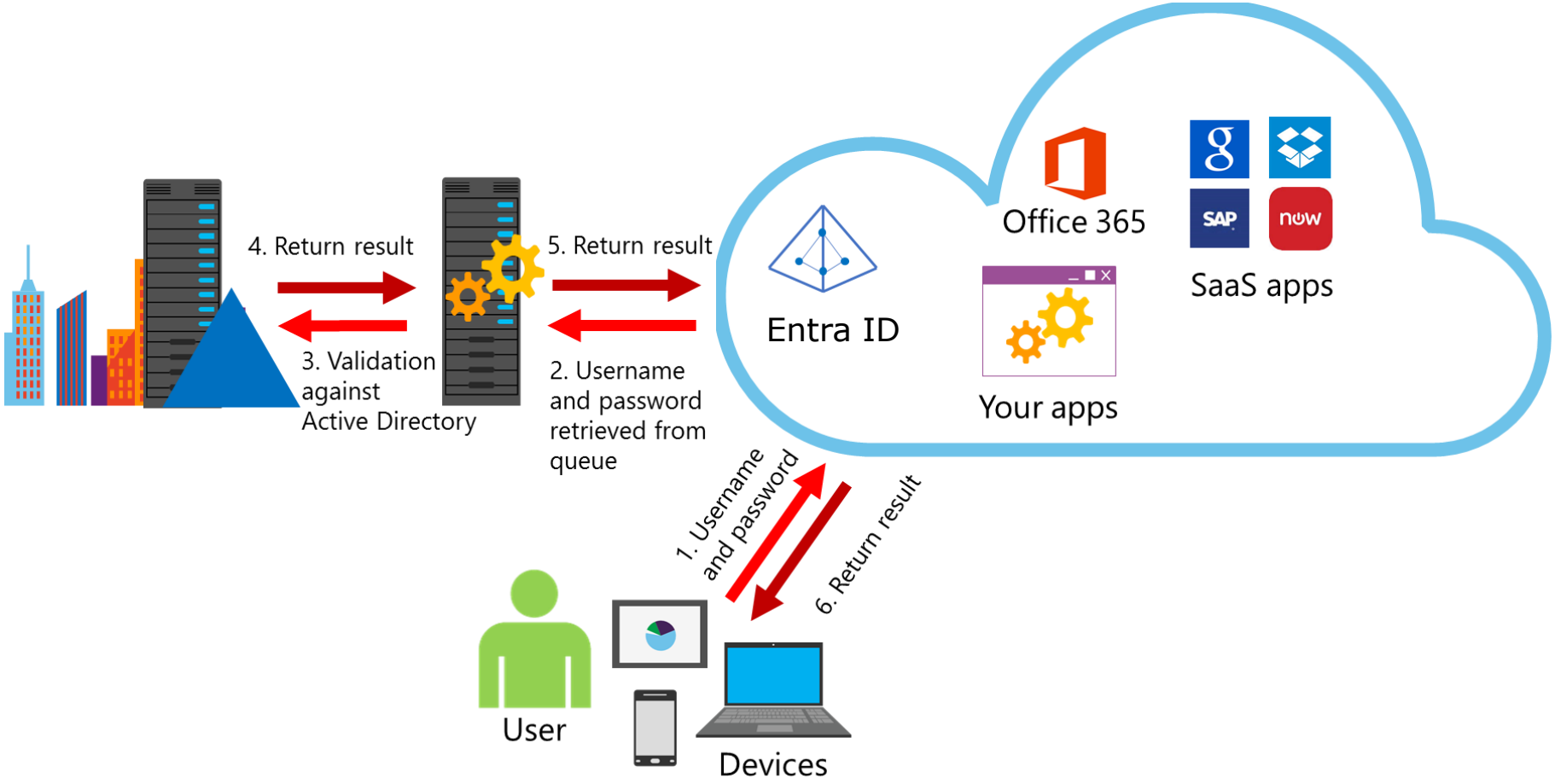
Entra Connect in multi-forest scenarios

- Entra Connect supports synchronization from mutiple AD DS forests
- You need to use a customized Entra Connect setup to configure this scenario
- Each user must be identified uniquely



Entra Connect pass-through authentication

Entra ID pass-through authentication enables you to centralize authentication for both on-premise and cloud resources without deploying AD FS



Lesson 3: Managing Microsoft 365 identities with directory synchronization

- Managing users with directory synchronization
- Managing groups with directory synchronization
- Modifying directory synchronization
- Monitoring directory synchronization
- Troubleshooting directory synchronization

Managing users with directory synchronization

After you deploy Entra Connect successfully and you enable scheduled synchronization, there are several required management tasks to ensure users are synchronize efficiently:

- Password writeback
- Device writeback
- Managing primary SMTP addresses
- Recovery from accidental deletes
- Recovery from unsynchronized deletes
- Accidental account deletion
- Bulk activation of new accounts

Managing groups with directory synchronization

- If deployed, the Exchange Server hybrid writeback is the classic writeback from Entra ID and is separate from a group writeback
- As such, an Exchange Server hybrid writeback is the only writeback option (for example, user writeback, password writeback, and device writeback) that does not require an Entra ID Premium license
- Otherwise, an Entra ID Premium P1 or P2 license is required if you enable a group writeback without the Exchange Server hybrid writeback feature

Modifying directory synchronization

The three filtering configuration types which can be applied to Entra Connect include:

- Domain, this filtering configuration type enables you to select which AD DS domains are allowed to synchronize to Entra ID, and uses Entra Connect or the Synchronization Service Manager
- OU, this filtering configuration type enables you to select which OUs in AD DS are allowed to synchronize to Entra ID, and uses Entra Connect or the Synchronization Service Manager
- Attribute, this filtering configuration type enables you to control which objects in AD DS should be synchronized to the Entra ID based on criteria of the object's attributes, and uses the Synchronization Rules Editor

Monitoring directory synchronization

Use these tools to monitor directory synchronization:

- Operations Manager, use the System Center Management Pack for Azure
- Microsoft 365 portal
- Classic and new Azure portal
- Event logs
- Windows PowerShell

Troubleshooting directory synchronization

- Key troubleshooting tasks for directory synchronization include analyzing logs for errors, and remediating synchronization errors with the tool
- Typical issues that can lead to problems include:
  - Installation errors, such as using incorrect on-premises or Microsoft 365 credentials
  - Inadvertently deactivating directory synchronization in the portal or through Windows PowerShell
  - Unexpected changes in AD DS that affect OU scoping or attribute filtering
  - Corrupted AD DS requiring directory recovery

Lab: Configuring directory synchronization

- Exercise 1: Preparing for directory synchronization
- Exercise 2: Configuring directory synchronization
- Exercise 3: Managing Active Directory users and groups

Logon Information

Virtual machines:	**20347B-LON-DC1** **20347B-LON-DS1** **20347B-LON-CL1** **20347B-LON-CL2**
User names:	**Adatum\Administrator** **Adatum\Beth** **LON-CL2\Francisco**
Password:	**Pa55w.rd**

Estimated Time: 90 minutes

Lab Scenario

The pilot deployment of Microsoft 365 is well underway at Adatum Corporation. The project steering committee has made the recommendation to continue with migrating additional departments to Microsoft 365. The first step in completing the migration is to configure directory synchronization so that user and group accounts will be synchronized for the on-premises AD DS domain rather than managing all user and group accounts in Microsoft 365.

Lab Review

- How do you configure OU level filtering for directory synchronization?

Module Review and Takeaways

- Review Question
- Real-world Issues and Scenarios
- Tools
- Best Practices
- Common Issues and Troubleshooting Tips